Melody Nguyen

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PROFILE

AI/Software engineer with experience in Python automation, machine learning pipelines, data processing, and interactive scientific tools. Strong foundation in algorithms, ML, and software engineering, with hands-on experience building scalable, reproducible systems at national labs.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, R, C, JavaScript (React)

- **ML/AI:** Scikit-learn, NumPy, Pandas, SciPy, Matplotlib
- **Web & Systems:** HTML/CSS, JavaScript, React, Flask, APIs, Web Services
- **Cloud & Databases:** AWS (EC2, S3, IAM), DynamoDB
- **Networking & Distributed Systems:** TCP/IP, DNS, Socket Programming, Distributed Systems
- **Software Engineering:** Agile, Scrum, Requirements, System Design, Testing, Project Management, AI
- **Tools:** Git, GitHub, Linux, Bash, Jupyter, Marimo, Tableau

PROFESSIONAL EXPERIENCE

Engineering Intern — Stanford, CA
Stanford Synchrotron Radiation Lightsource (SSRL) — *Jun 2025 – Aug 2025*

- Built Python automation to streamline X-ray diffraction data collection across 130+ runs
- Developed ML-based peak detection models and scalable pipelines for high-volume scientific data

Product Ambassador & Software Contributor — New York, NY
marimo.io — *Sep 2024 – Dec 2024*

- Developed interactive Python notebooks demonstrating reproducible scientific computing
- Contributed code, documentation, and UI/UX feedback to enhance platform interactivity
- Authored tutorials in Python data visualization, increasing adoption among researchers

Data Science Intern → Software & Web Developer — Stanford, CA
SLAC National Accelerator Laboratory, U.S. Dept. of Energy — *Mar 2023 – Aug 2024*

- Analyzed 60+ photovoltaic datasets using Python, Pandas, and ML techniques; integrated NOAA + NREL data into automated pipelines for multi-state climate modeling
- Built visualizations and geospatial tools to support renewable grid resilience research
- Developed and maintained 5+ grid-simulation platforms (HTML/CSS, JavaScript, Jekyll)
- Wrote documentation and onboarding guides for cross-functional engineering teams

PROJECTS & RESEARCH

Skill Sync — React, Scrum — Software Engineering Capstone — *2026*

- Barter platform to match users and facilitate skill exchange; designed full-stack system with team using Agile

Extreme Weather + Solar Performance Modeling — Python — *2024*

- Merged NOAA + NREL datasets to quantify grid stress events
- Visualized cross-regional performance losses during heatwaves

EDUCATION

Pace University, Seidenberg School of Computer Science and Information Systems — New York, NY
Bachelor of Arts (BA) in Computer Science | **Minor**: Business | Dean's List |**GPA**: 3.81 — *May 2026*

- **Honors Thesis**: Digital Pollution: Understanding Energy Costs of Large Language Models
- **Relevant Coursework**: Artificial Intelligence, Software Engineering, Machine Learning, Data Science, Algorithms, Networks and Internet, Object-Oriented Programming

Georgia Institute of Technology — Online
Masters of Science, Computer Science — *Incoming Fall 2026*

Cornell University, Product Management Certificate — New York, NY
Product hypothesis & personas, roadmap, prototyping, analytics, engineering and execution — *Feb 2026*